

SOFTWARE ENGINEERING — PROJECT PROPOSAL

NextHire

An AI-Powered Resume Filtering and Recruitment Platform

Prepared by

Team Innovex

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1. Team Formation

NextHire is being developed by Team Innovex, a group of five students from the Department of Computer Science and Engineering. The team was formed based on complementary technical skills, with each member taking responsibility for a specific area of the project. This structure ensures clear ownership and avoids overlap during development.

Team Name: Innovex

Project Name: NextHire

#	Name	Role
01	Afsana	Team Leader : Leadership, communication, project management (Trello), documentation, Git.
02	Sabiha Akter	Database Administrator : Database design (ERD, normalization), SQL optimization, data security, MySQL, PHP, Python.
03	Tasnim Ahmed Arnob	Backend Engineer : Requirement gathering, stakeholder analysis, PHP(Laravel), NPL (spaCy)
04	Mahbubur Rahman Antor	UI/UX Designer and Fronted Developer : Figma, Adobe XD, user research, responsive design, HTML, CSS, JavaScript, Bootstrap.
05	Shouvik Biswas	SQA Engineer & Business Analyst : Test planning, system testing, bug tracking, QA Tools, Market research, competitor analysis, business model, reporting.

Table 1: Team Innovex — Member Roles

2. SWOT Analysis of Team

Before starting development, the team conducted a SWOT analysis to understand its strengths and identify areas that may need extra attention during the project.

Strengths • Combined skills covering full-stack and database development • Clear understanding of the recruitment problem domain • Each member has a focused, non-overlapping responsibility • Familiarity with REST APIs and relational databases	Weaknesses • Limited hands-on experience with large-scale AI model deployment • Short project timeline given the scope of features planned • No existing user base or company partner for real-world testing
Opportunities • Growing demand for HR automation tools among small businesses • Freely available open-source NLP libraries reduce development cost • Potential to extend the project into a commercial product later • Addresses a real, measurable pain point in hiring workflows	Threats • Established HR platforms with large existing user bases • AI output quality depends heavily on training data availability • Data privacy regulations may affect how candidate data is stored

Table 2: SWOT Analysis — Team Innovex

3. Idea Generation

The team explored several problem areas before settling on recruitment as the project domain. During initial discussions, it became clear that most organizations handle hiring in a very manual and time-consuming way, which creates unnecessary delays and inconsistencies in the selection process.

3.1 Domain and Problem Identification

We focused on the Human Resources and recruitment domain after observing how companies — particularly smaller ones — struggle to handle a high volume of job applications. Without automated tools, HR teams end up spending most of their time just reading CVs rather than actually evaluating candidates. This is a practical, well-defined problem that technology can meaningfully address.

3.2 Why NextHire Was Selected

- **Feasibility:** The project can be realistically completed within one semester using tools the team already knows.
- **Impact:** It solves a real problem that affects businesses of all sizes, not just large corporations.
- **Innovation:** Adding AI-based resume matching and summarization goes beyond what most basic recruitment tools offer.
- **Scope:** The core system can be built as a working prototype, with room to add more features later.
- **Learning Value:** The project covers frontend, backend, database, and AI development — valuable across all team members' academic goals.

4. Introduction and Overview

NextHire is a web-based recruitment management platform designed to help companies handle the hiring process more efficiently. The system allows organizations to post job requirements, collect applications, and automatically screen and rank candidates using AI-based resume analysis.

The platform is aimed at HR teams, recruitment managers, and any organization that regularly handles a large number of job applications. The core idea is straightforward: instead of manually going through each CV, the system reads and evaluates resumes on behalf of the recruiter, assigns a match score, and presents the best candidates at the top of the list.

NextHire also supports multiple companies within the same platform. Each organization gets its own separate dashboard, job posts, and applicant pool. This makes it suitable not only for individual companies but also for institutions or agencies managing recruitment for multiple clients.

The expected outcome is a meaningful reduction in the time it takes to move from receiving applications to scheduling interviews. By automating the most repetitive part of the recruitment process, HR teams can spend their time on decisions that actually require human judgment, such as conducting interviews and evaluating cultural fit.

5. Problem Statement

Despite advances in technology across almost every business function, recruitment in most organizations is still a largely manual process. HR teams receive applications by email or through basic job portals, then read through each CV individually to decide who moves forward. When a job post attracts hundreds of applications, this quickly becomes unmanageable.

The problems with this approach are well-documented and widely experienced:

- **Volume:** A single job opening can receive hundreds of CVs within days. Reading each one properly is simply not practical for a small HR team.
- **Inconsistency:** Different reviewers apply different standards when reading resumes, leading to uneven evaluations across the same applicant pool.
- **Slow process:** When screening is manual, the time between application submission and the first interview call can stretch into weeks, causing strong candidates to accept offers elsewhere.
- **No ranking system:** Without a structured way to compare candidates, decision-

making becomes subjective and harder to justify.

- **Poor scalability:** Tools like spreadsheets and email inboxes were not designed for managing recruitment at scale, yet many organizations still rely on them.

These issues are not just operational inconveniences — they have a direct effect on the quality of hiring decisions and the ability of a company to attract and secure the right people. NextHire is designed to address these problems directly with a system that is practical, accessible, and built for real-world use.

6. Motivation

The idea for NextHire came from a straightforward observation: most of the companies and institutions around us still rely on manual processes to handle recruitment. Having spoken with people who work in HR roles, or who have applied to jobs and waited weeks without a response, it was clear that the process is frustrating from both sides.

What motivated the team most was the realization that this is a solvable problem. The technology needed to build a smarter hiring system — from basic web development to NLP-based text analysis — is accessible to a group of students working on a semester project. There is no reason for companies to still be reading through hundreds of CVs by hand when a well-built tool can do the initial screening accurately and consistently.

From a broader perspective, improving recruitment has real value for society. When companies hire more effectively, they find better candidates. When the process is faster and more transparent, applicants have a better experience and are less likely to be overlooked due to manual error. For small businesses in particular, a tool like NextHire could make a significant difference.

For the team, this project is also an opportunity to build something that works end-to-end: a real system with a proper backend, a database, a usable interface, and an AI component. That combination makes it one of the most complete and meaningful projects we could work on at this stage of our studies.

7. Expected Features and Requirements

The following table outlines the core features of the NextHire platform. Each feature has been chosen to address a specific part of the recruitment workflow identified in the problem statement.

Feature	Description
Resume Storage & Search	Stores all submitted CVs in a structured database, searchable by skill, education level, and experience.
AI Resume Filtering	Automatically screens resumes against job criteria using NLP, reducing manual reading time significantly.
Candidate Match Score	Assigns each applicant a percentage score based on how closely their profile matches the posted job requirements.
Applicant Ranking	Sorts candidates from best to least suitable so recruiters can focus on the most relevant profiles first.
Smart Shortlisting	Allows companies to set an interview capacity limit; the system selects the top candidates within that number.
AI Resume Summary	Generates a short, readable summary of each CV so recruiters can review key points without reading the full document.
Multi-Company Support	Each company gets its own dashboard, job posts, and recruitment data — fully separated from other organizations.
Job Post Management	Companies can define the skills, experience, and qualifications required for each role they advertise.

Feature	Description
Recruitment Dashboard	A central overview showing total applications, shortlisted candidates, and the current stage of each hiring process.
Internal Messaging	A built-in chat tool for HR team members to discuss candidate evaluations without leaving the platform.
Feedback System	Companies can rate the platform and leave suggestions, helping improve accuracy and usability over time.
Accepted Candidate Table	Displays shortlisted and accepted applicants in a clean, organized table format for easy tracking and reference.

Table 3: NextHire — Core Features and Requirements

8. Development Approach

8.1 Development Model

NextHire will be developed using the **Agile** model. This approach was chosen because it allows the team to build and test features progressively, rather than trying to plan and deliver everything at once. Work will be organized into two-week sprints, with each sprint producing a testable increment of the system. Regular reviews at the end of each sprint will allow the team to adjust priorities based on what is working and what needs improvement.

8.2 Tools and Technology Stack

Component	Technology
Application Type	Web application, browser-based, fully responsive
Frontend	React.js with Tailwind CSS,javascript,Bootstrap,HTML5
Backend	Node.js with Express.js — REST API architecture,PHP(Laravel)
Database	MySQL for structured data storage
AI / NLP	Python microservice using HuggingFace Transformers or OpenAI API or spaCy
Authentication	JWT-based login with role-based access control
Version Control	Git with GitHub — all members added as collaborators
Testing	PHPUnit
Deployment	ExonHost cpanel(Bangladesh)
Management	Trello
Security	SSL + password_hash()
Design Tool	Figma for wireframes and UI prototypes

Table 4: Technology Stack

9. Budget Estimation

The figures below represent a realistic estimate for developing and running NextHire within the scope of this academic project. Most development tools are free, with costs concentrated in hosting and API usage.

Category	Details	Estimated Cost(BDT)
Development Tools	VS Code, GitHub, Postman	Free
Cloud Hosting	VPS or cloud server for 6 months	80,000 – 1,20,000
AI / NLP API	OpenAI or HuggingFace (usage-based)	1,00,000 – 1,50,000
Domain & SSL	Custom domain with HTTPS certificate	15,000 – 25,000
Database Hosting	PostgreSQL cloud (Supabase free tier)	Free / 40,000 – 60,000
UI Libraries	Tailwind CSS, Figma	20,000 – 40,000
Testing	System testing, deployment	30,000 – 50,000
Maintenance	Server maintenance, backup	50,000 – 70,000
Miscellaneous	Testing, printing, documentation	20,000 – 40,000
Total		4,00,000 – 5,50,000

Table 5: Budget Estimation

10. Project Timeline

The project will run over twelve weeks, following the phases of the Software Development Life Cycle. The frontend and backend phases overlap intentionally, allowing parallel work and more efficient use of the available time.

Phase	Activity	Duration	Description
Phase 1	Requirements & Planning	Week 1–2	Define scope, finalize requirements, assign tasks, set up the GitHub repository.
Phase 2	UI/UX Design	Week 3–4	Create wireframes and mockups in Figma. Design the database schema and system architecture.
Phase 3	Backend Development	Week 5–7	Build REST API endpoints, implement authentication, and integrate the AI resume filtering service.
Phase 4	Frontend Development	Week 5–8	Develop the React-based interface, connect to the backend, and build the recruiter dashboard.
Phase 5	Database Integration	Week 6–8	Set up PostgreSQL, define relationships, and optimize queries for resume storage and retrieval.
Phase 6	Testing & QA	Week 9–10	Conduct unit and integration testing. Fix bugs and validate AI scoring accuracy with sample data.
Phase 7	Deployment & Handover	Week 11–12	Deploy to cloud server, finalize documentation, and prepare the presentation and final report.

Table 6: Project Timeline — 12-Week SDLC Plan

11. Conclusion

NextHire addresses a genuine problem that organizations of all sizes face when managing recruitment. Manual CV screening is slow, inconsistent, and difficult to

scale — and there is a real need for tools that can handle this process more reliably. This project aims to deliver exactly that: a working recruitment platform that uses AI to take over the most repetitive parts of hiring, so that HR teams can focus their attention where it actually matters.

The team has the right mix of skills to build this system, with each member covering a distinct and necessary part of the development work. The technology choices are practical and well within what a student team can realistically implement and test within a semester.

Beyond the academic scope, NextHire has the potential to grow into a product that could genuinely be useful to small businesses and institutions that cannot afford enterprise-level HR software. The project is both technically ambitious and practically grounded, which is why the team is confident it will result in a system worth submitting, presenting, and potentially building on further after the course concludes.